

A smart Healthcare system manages patients, doctors, appointments and medical records.

Explain how these concepts improve such as class, object, encapsulation, inheritance, and polymorphism are applied in this system.

① Class

A class is a blueprint of an object.

In a smart Healthcare system, classes can include

- * Patient
- * Doctor
- * Appointment
- * Medical Records

example:-

→ patient class contains attributes such as patient ID, name, age and contact number.

→ It contains methods such as bookAppointment() and viewMedicalRecord().

② Object

An object is a instance of class

example:-

patient: John Smith

Doctor: Dr. Emily Brown

Appointment: Appointment #101

Each object has its own data while sharing the behavior defined by its class.

③ Encapsulation

Encapsulation means hiding sensitive data & allowing access only through public methods.

example :-

patient medical history is declared as private.

It can only be accessed using methods like

- getMedicalHistory()
- updateMedicalHistory()

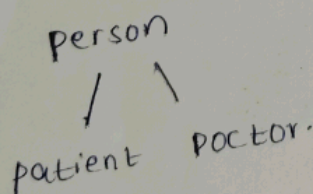
Benefits

- protects confidential patient information.
- prevents unauthorized modification.
- Improves data security and integrity.

④ Inheritance

Inheritance allows one class to inherit properties and methods from another class.

ex:-



person class contains common attributes

- Name
- Age
- Address

→ patient & doctor inherit these attributes and their own features.

Benefits

Reduces code duplication

promotes code reuse

makes future enhancements easier.

polymorphism

polymorphism allows the same method to perform different actions depending on the object.

ex

method

displayDetails()

Analysis of Benefits

① Scalability

- New classes such as Nurse, pharmacist, or laboratory Technician can be added easily using inheritance
- New features such as online consultation or e-prescriptions can be integrated with minimal changes to existing code.

② Security

- encapsulation ensures sensitive patient data is hidden from unauthorized access.
- Access can be controlled through methods and user roles, helping maintain confidentiality & data integrity.

Maintainability.

→ classes separate responsibilities making the system modular.

→ Reusable code through inheritance reduces duplication

object-oriented programming provides a structured and efficient way to develop a smart health care system.